Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Lillian Emerson	lilyeme@udel.edu
Friend (optional)	Shreya Anthey	Shreya Anthey

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

To create some sort of website that raises awareness about more unknown environmental issues in a fun and engaging manner, maybe habitat loss centered and users can pick an animal (that has struggled with habitat loss at human hands) they want to be and "experience" different scenarios the animals face due to it
To compile a list of ways to start being more environmentally conscious (little swaps of everyday things or something not many people consider) for people who are interested in an eco-friendly lifestyle but overwhelmed and don't know where to start (maybe based on a quiz users take?)
The water game - follow water through treatment and where it goes after the drain and whatnot

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are people on social media who work in water treatment and post about it, and websites describing how water is treated and returned.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I hope the website helps bring awareness to environmental related topics and makes people think critically about the world we live in, and hopefully be used to educate people about water and how important it is and how it gets used and recycled.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

the ultimate goal is to walk the user through the water cycle and how water gets treated after use based off of users input and what they want to see

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

they will most likely have to provide what type of water they want to start with or possibly where (what city) they want to start at to see how water gets processed and returned

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

using if statements the computer will determine what path to take through the water treatment cycle and give the user interesting or fun facts about the water cycle on the way, also using loops to filter through user input at certian parts

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

the user will get back a tour of the water treatment cycle in the general area they selected in the beginning with fun facts along the way

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Water, Location, Treatment stage,

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need- Water (what utility/industry it came from, how it will be treated, what will be done with it afterwards) *details i will ignore- Water (location, amount, cost of treatment)*

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I think the most complex part will be organizing the different paths the water treatment can go and determining how to give the options in a not confusing manner to the user

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will work on this project the weekends before it is due and get a reasonable chunk of it done and then also wednesdays are my most free day so around 9am i will work on it for an hour or 2 and finish up what i need to get done for the week.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal for this project is to learn more about the water treatment cycle because its something im interested in and also a possible career i want to explore, and i would also like to improve my drafter skills.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to make a fun interactive web tool that teaches people about the water treatment cycle. what are 3 things that you already know about the water treatment cycle, and what are 3 things you would like to know about it?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

my friend suggested that i incorporate the different types of treatment the water can go through, and she was interested in what happens with the water after it gets treated and how it is transported and stored or returned, she didnt really mention existing solutions or prior apps other than people who work in water treatment on social media explaining what they do and bringing awareness to it.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I think that explaining multiple different types of treatment would be overly ambitious since im already tracking the water from its general location to the treatment center and then

where it goes after, and that would be a lot of code and very confusing to manage and organize

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My project idea has become more focused by narrowing down the possible pathways the water can take through treatment

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I plan on going to office hours if i get stuck and getting help there, and also examples of code from previous lessons and modules in this course.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I do not plan on using an LLM because i dont appreciate the environmental impacts it has. when i get an error in my code i will step through my code line by line and use other methods described in this course.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.